How accurate LIGHT is

The basic network of LIGHT v0, version 0.1.0, on 28 September 2026, with its energy ladder, its system map and its reaction rates

Bottom line. LIGHT v0 reproduces the fitted KrF decays [1, 2] within 2.3 to 6.1 % rms and predicts decays that no fit used [4] within 11.7 % rms. At the Tahoma laser [5] its default network, which holds n_e fixed and has no charge chemistry, gives 32 % of the measured small-signal gain of 6.39 %/cm, and with its optional ion, F_2-electron and Kr_2^* groups on and n_e free, as ML scores it, 96 %.

That configuration also gives 110 % of Tahoma's saturation intensity and 105 % of its formation efficiency, and the Tahoma scores in Table 1 use it. v0_1 and v0_2 carry the charge chemistry in their default networks. The C/B emission ratio of Yu and Setser [3] is matched to -0.5 %. No fit used Tahoma's numbers and no rate was set by a laser measurement, so the Tahoma scores are predictions.

Apart from the default network's gain at Tahoma, its largest misses are on data that conflict with the fitted sets, 19.3 and 34.1 % rms on the C-A decays of Hakuta et al. [6] and -22.3 % on the KrF(B) lifetime of Burnham and Searles [7], and the VUV yield of pure krypton at low pump, 136 % of Morozov's measurement [8].

[1] Eden, Waynant, Searles and Burnham, Appl. Phys. Lett. 32, 733 (1978). [2] Gadomski, Xu, Setser and Nelson, Chem. Phys. Lett. 189, 153 (1992). [3] Yu and Setser, J. Phys. Chem. 94, 2934 (1990). [4] Xu, Gadomski and Setser, J. Chem. Phys. 99, 2591 (1993). [5] Spectra Technology, *Experimental Support for KrF Laser Modeling*, final report to Los Alamos, 19 July 1989, OSTI 10136715. [6] Hakuta et al., Jpn. J. Appl. Phys. 29, 1194 (1990). [7] Burnham and Searles, J. Chem. Phys. 67, 5967 (1977). [8] Morozov, Heindl, Krücken, Ulrich and Wieser, J. Appl. Phys. 103, 103301 (2008).

LIGHT is a 0-D model of krypton and fluorine under a power drive. Its basic network has 24 reaction rows. Every KrF rate among them but one is a measurement of that process or a joint fit to kinetic measurements of it. The exception is the KrF(B) radiative lifetime, where LIGHT keeps Kannari's 6.99 ns and Eden measured 6.8 ± 0.2 ns. No rate was set by a laser measurement. The one independent Kr/F_2 laser LIGHT has been scored against is Tahoma, and no fit used any number from it. There LIGHT gives the small-signal gain at $\times 0.96$ the measured value, the saturation intensity at $\times 1.10$ and the formation efficiency $g_0 I_{sat}/P_{in}$ at $\times 1.05$. The largest open questions are three links of the formation chain that no one has measured and the trimer rate above 600 K. Table 1 lists every test.

1 How the numbers were made

Each dataset has one role. Kinetic measurements of the channels being fitted train them. Other kinetic measurements are held out and test the fit. Laser performance only scores, and the fitting code refuses a dataset of that kind. The train set is the largest group of kinetic datasets that agree with one another within their errors, found by leaving each group out in turn. This is the repository's predictive-only policy. Measurements score the model after the fact and never steer a number. Every number in this note is written by `docs/make_overview.py` from LIGHT and from `ML/`, and a test fails when those files no longer match the model.

2 The KrF(B,C) block

KrF has two emitting states. B lases at 248 nm, and C radiates slowly in a band near 275 nm. LIGHT treats them as the two coupled levels of eq. 5 of Gadomski et al. (1992) and reproduces that model's analytic decay rate to within 1.0×10^{-4} at every pressure tested. The block has 10 constants. Nine come from one joint fit to three measurement sets that agree with one another, the KrF(B) decays of Eden et al. (1978), the Kansas State B/C decays (Gadomski et al. 1992) and Yu and Setser's equilibrium C/B

test	measured	LIGHT	role of the data
energy ledger closure, 9 runs across the modes		below 10^{-12} of E_{dep}	bookkeeping
KrF(B) decay against Kr and F_2, Eden 1978		rms 2.3 and 3.2 %	train
KrF(B,C) slow decay, Gadomski 1992 Figs. 1 and 2		rms 6.1 and 4.2 %	train
C/B emission ratio at 300 K, Yu and Setser 1990	0.16 ± 0.02	-0.5 %	train
KrF(B,C) decays, Xu 1993 Figs. 5 to 7		rms 11.7 %	held out
decay constants Xu 1993 printed, 50 and 100 Torr Kr	5.0×10^{7} and 6.2×10^{7} s^{-1}	$\times0.91$ and $\times0.97$	structure check
C-A decays, Hakuta 1990		rms 19.3 and 34.1 %	held out, in conflict
τ_B, Burnham and Searles 1977	9.0 ± 0.6 ns	-22.3 %	held out, in conflict
C/B ratio 230 K over 300 K, Yu and Setser	1.30 ± 0.06	-13.6 %	held out
Kr^* + Kr^+ made per eV, pure Kr and Ar, Saito 2003		$\times1.02$ and $\times1.03$	check
Kr_2^+ + e^- recombination, Shiu and Biondi 1977		$\times0.87$ to $\times1.08$	check
$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$, Smith et al. 1972	2.74×10^{-31} to 3.3×10^{-31}	2.5×10^{-31}	check
VUV yield of pure Kr at low pump, Morozov 2008	0.42 ± 0.05	0.57, $\times1.36$	check
electron closure at Tahoma, kinetic EEDF loop	T_e 1.61 eV	1.59 eV	reference
Tahoma small-signal gain g_0	6.39 %/cm	6.11, $\times0.96$	device, score only
Tahoma saturation intensity I_{sat}	1.4 ± 0.1 MW/cm^2	1.54, $\times1.10$	device, score only
Tahoma formation efficiency $g_0 I_{\mathrm{sat}}/P_{\mathrm{in}}$	0.27	0.284, $\times1.05$	device, score only

Table 1: LIGHT against the data. Rates are in cm^{3}s^{-1} or cm^{6}s^{-1}. A percentage is LIGHT's deviation from the data, signed for a single value and rms for a set. A $\times$ factor is LIGHT over the measurement.

ratio (1990). The tenth, τ_B, stays at Kannari's 6.99 ns, inside the fit's interval of 6.79 to 7.22 ns. The fit's median and 68 % interval for B-C mixing by Kr is $k_{BC} = 9.8\times10^{-12}$ [4.63×10^{-12}, 1.73×10^{-11}] cm^{3}s^{-1}. The equilibrium ratio is $K_{\mathrm{eq}} = 1.07$ [0.84, 1.38], and the C lifetime is $\tau_C = 50$ [41, 63] ns. LIGHT carries the best fit, $k_{BC} = 1.05\times10^{-11}$ cm^{3}s^{-1}, $K_{\mathrm{eq}} = 1.10$ and $\tau_C = 50$ ns. Mixing is slow, so KrF made in C mostly stays there. At 1 atm and 300 K the lasing B state holds 0.69 [0.57, 0.80] of the manifold, and LIGHT's rounded constants give 0.68.

The fitted decay sets are matched to between 2.3 and 6.1 % rms. The held-out decays of Xu et al. (1993) are matched to 11.7 %, which is inside the 30 % by which the Kansas State figures scatter at one condition. The two decay constants Xu et al. print come out at $\times0.91$ and $\times0.97$, where the notes' list as written, with no C state, gives $\times2.89$ and $\times2.41$. Three held-out measurements disagree. Hakuta et al. (1990) measured faster C-A decays, missed by 19.3 and 34.1 % rms. Their data also conflict with the three fitted sets, which is why they are held out. Burnham and Searles (1977) measured $\tau_B = 9.0$ ns, where LIGHT uses Kannari's 6.99 ns and Eden measured 6.8 ns. Yu and Setser's C/B ratio between 230 and 300 K comes out -13.6 %, because the energy ledger books a B-C gap of 80 cm^{-1} where their two ratios imply 180 ± 30 cm^{-1}.

Tahoma's gain is the sharpest test of the block, since g_0 is the KrF(B) density times the stimulated cross section. Over 40 draws from the fit's posterior the gain is 5.83 %/cm [5.42, 6.63] against the measured 6.39.

3 KrF formation

The chain from the beam's Kr^+ and Kr^* to KrF sets the formation efficiency. LIGHT carries every link that has been measured at its measured value. $Kr^* + F_2$ reacts at $8.47\times10^{-10}\,cm^3s^{-1}$, the mean of three first-hand measurements. Velazco, Kolts and Setser (1976) measured the KrF^* yield of that reaction as 90 %, the KrF^* formation rate constant over the total quenching, and published it as unity $\pm$ 0.20. Yu and Setser (1990) put 38 % of that KrF^* in C. The Kr_2^* route carries Ueda's Kr^* + 2Kr rate, the measured 270 ns Kr_2^* lifetime and Quigley and Hughes' $Kr_2^* + F_2$ rate. Appendix A gives every value with its source.

Tahoma measured $g_0 I_{sat}/P_{in}$ = 0.27. ML puts a 248-nm probe flux on LIGHT and fits its I_{sat} at fixed g_0, as Tahoma did. That gives 0.284. Its KrF formation rate in 5-eV quanta is 0.318 of the pump. The difference is KrF that forms in C, or mixes into it, and decays there without meeting the laser field.

Table 2 moves one link at a time to another value it could take. The measured range of the harpoon yield, 0.70 to 1.0, moves $g_0 I_{sat}/P_{in}$ from 0.256 to 0.298 and brackets Tahoma's 0.27. Three links have no measurement at all. They are the KrF^* yield of ion-ion recombination with F^-, the share of that KrF formed in C, and $Kr_2^* + F \rightarrow KrF$. Only analogs exist for the first two, and both analogs lower $g_0 I_{sat}/P_{in}$ toward the measurement. Removing the third lowers it too. The predictive-only policy lets no analog set a link and no link be chosen by its fit to Tahoma. The rest of the Kr_2^* route moves $g_0 I_{sat}/P_{in}$ by 0.006 at most.

link	LIGHT	tried	what the value is	$g_0 I_{sat}/P_{in}$	change
KrF^* yield of $Kr^* + F_2$	0.90	0.70	the measured 0.90 less its stated 0.20	0.256	−0.028
		1.0	the published assignment	0.298	+0.014
		0.5	as for $Ar^* + F_2$	0.227	−0.057
share of that KrF formed in C	0.38	0	if nascent C relaxes to B at 1 atm	0.307	+0.023
KrF^* yield of ion-ion recombination	1	0.64	as for Cl^-	0.241	−0.043
share of that KrF formed in C	0	0.38	as for SF_6^-	0.266	−0.018
ion-ion recombination rate	Flannery	×0.38	Avco's model input	0.284	−0.001
$Kr_2^* + F \rightarrow KrF(B) + Kr$	3.0×10^{-10}	0	no measurement exists	0.268	−0.017

Table 2: Each formation link moved alone from LIGHT's chain at Tahoma, 2.8 Torr of F_2 in Kr at 665 Torr in all, pumped at 330 kW cm^{-3}. LIGHT's chain gives 0.284, and Tahoma measured 0.27.

4 The electron side

LIGHT integrates each krypton electron-impact rate over an EEDF with two temperatures. The bulk is a Maxwellian at T_e. Above the 9.915-eV threshold the tail has its own temperature, read from a table of kinetic EEDFs with electron-electron collisions at e-beam-sustained states. Each electron that is captured takes the energy it really carries, $\langle\varepsilon\sigma v\rangle/\langle\sigma v\rangle$. For F_2 attachment at Tahoma that is 0.55 eV, where $\frac{3}{2}T_e$ would book 2.40 eV. A kinetic loop that iterates LIGHT with a Boltzmann solution is the reference. At Tahoma LIGHT's own closure lands within 0.7 % of the loop's T_e and 2.3 % of its n_e, and within 1.0 and 0.5 % of its gain and formation.

The cross sections were checked against swarm data. With Biagi's krypton set the computed W is 0.96 to 1.03 times Nakamura's measured values. For F_2 attachment the rate constants measured by different groups disagree beyond their errors, and the SIGLO set LIGHT uses gives between 0.48 and 2.41 times the measured values. The two groups that agree give a scale of 0.92 $\pm$ 0.04 on the main resonance, 1.8 standard deviations from SIGLO. LIGHT keeps SIGLO as it is.

On pure gases the fluorine-free half of the chain checks out. The beam drive's Spencer-Fano yields make 1.02 times the Kr^* and Kr^+ per eV that the measured W and N_{ex}/N_i give in Kr, and 1.03 times

in Ar. LIGHT's Kr_2^+ recombination is 0.87 to 1.08 times Shiu and Biondi's between 0.026 and 2 eV. The VUV yield of pure Kr at low pump is the one miss. LIGHT gives 0.57 where Morozov et al. (2008) measured 0.42 $\pm$ 0.05. Without the thermal electrons' excitation LIGHT gives 0.55, close to the one photon per excitation and per ion that the measured W and N_{ex}/N_i allow, 0.54. No rate in LIGHT removes the rest. The measured e-beam yields of Ne, Ar, Kr and Xe all fall below that ideal, so the shortfall is either physics LIGHT lacks or a systematic error of the measurement. In pure Kr at low pump the bulk EEDF is not Maxwellian either, and the one-temperature energy equation cannot follow it there. That limits the pure-gas check and not KrF.

5 The energy ladder

Figure 1 places every species of v0 at its formation energy, the energy above ground-state Kr and F_2 at rest that Appendix A lists, and draws the rows that move energy between them. It follows the ladder view of the website's excimer simulation, with v0's own numbers. v0 keeps REBUILD's estimated KrF(B), 6.15 eV, which puts it 5.25 eV below the ion pair $Kr^+ + F^-$ and 5.35 eV above Kr + F. The 248-nm photon carries 5.00 eV of that and leaves KrF(X) 0.35 eV above Kr + F, which heats the gas as X falls apart. The harpoon G3 heats the gas by 2.97 eV for each KrF(B) it makes and the trimer's formation by 2.05 eV. Kr_2F^* emits 3.00 eV at $\sim$415 nm and leaves 0.30 eV. The Kr_2^* route, most of the charge chemistry and the F_2 electron rows are optional groups in v0, drawn dashed, and the species only they make are pale.

6 The system map

The map on the next page draws the basic network's energy flow at the notes' reference point, 1 atm of Kr with 0.5 % F_2, 300 kW cm^{-3} for 1 μs and $n_e = 10^{15}$ cm^{-3} held. Band widths are proportional to the energy they carry, and every box conserves it. Of the deposited energy, 28.0 % passes through KrF in 5-eV quanta. Of that KrF, 18.7 % radiates at 248 nm, 50.5 % forms the trimer and 19.0 % is quenched by electrons, so 5.2 % of the deposited energy leaves at 248 nm. The notes' six reactions as written, drawn in `docs/system_map.pdf`, give 16.8 %. The difference is the C state and the collisional losses that the KrF measurements require.

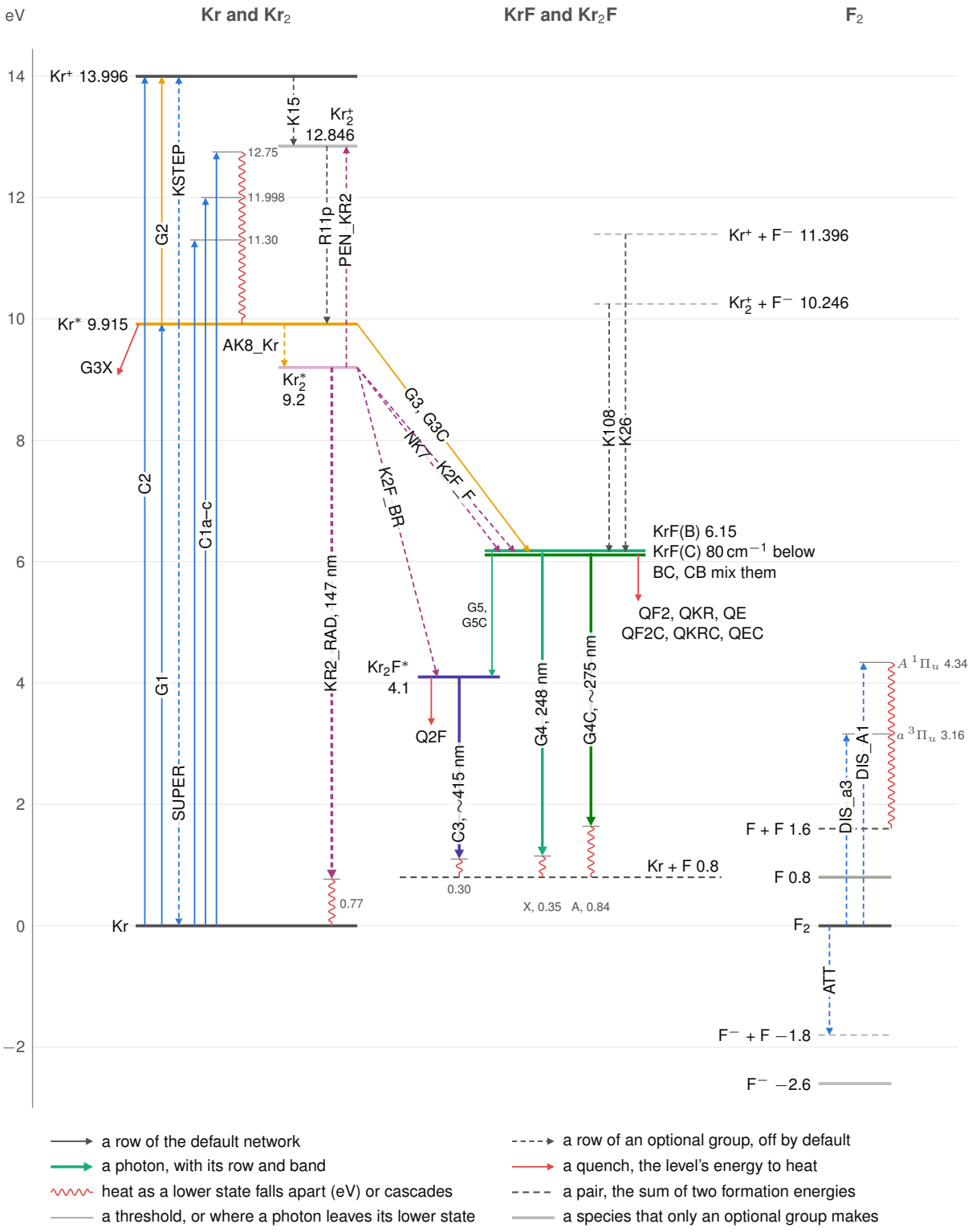
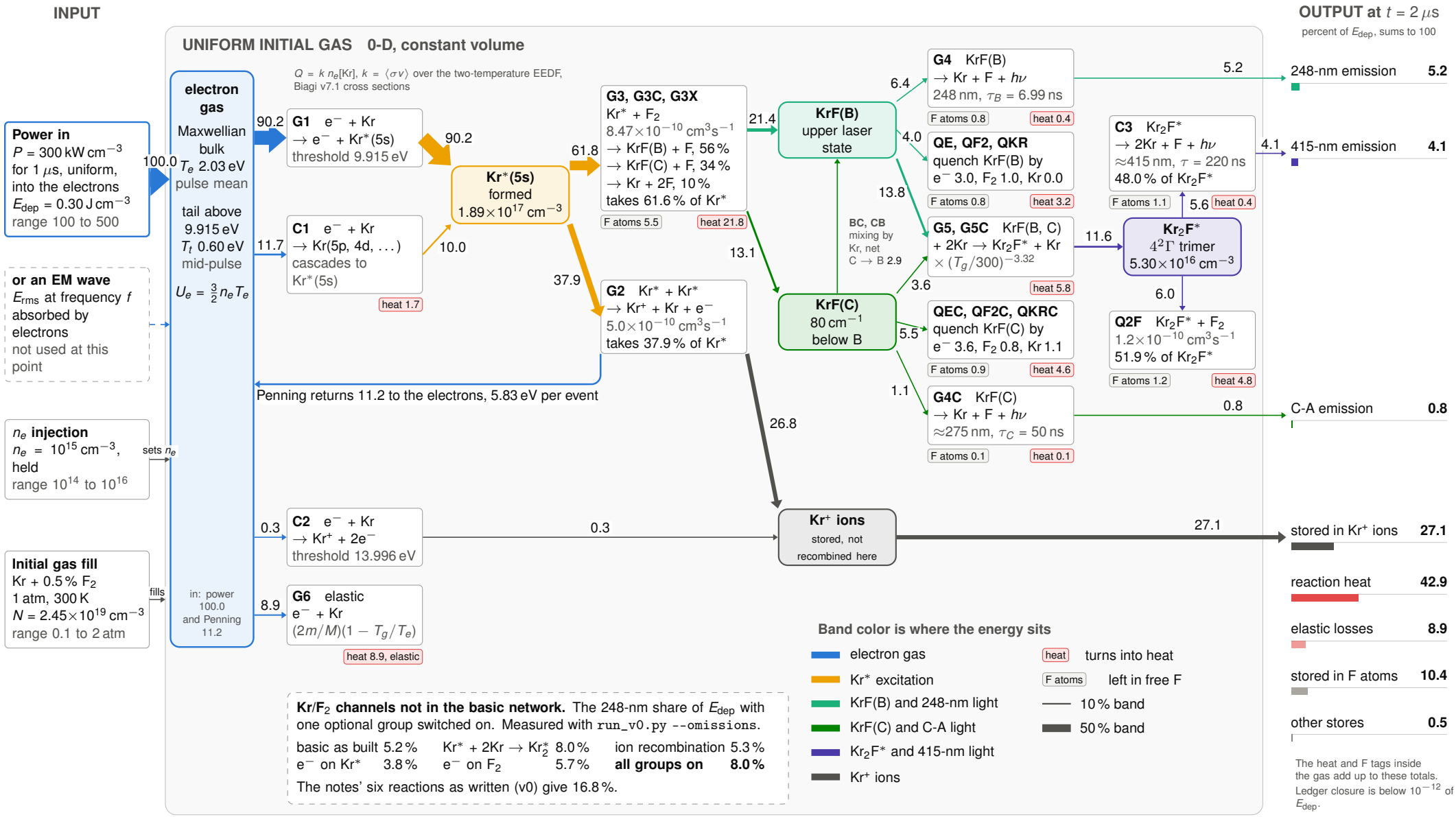


Figure 1: The energy ladder of LIGHT v0, version 0.1.0, in eV above ground-state Kr and F_2 at rest. A bar is a species at its formation energy and a dashed bar a pair, the sum of two. An arrow is a row, labelled with its id and solid in the default network. The rows of the optional groups `kr2`, `ions`, `electron` and `f2e` are dashed, and the species only they make, Kr_2^*, Kr_2^+ and F^-, are pale. A photon's arrow drops by its quantum, and the red squiggle below it is the heat its lower state gives the gas as it falls apart. A red stub is a quench. The energy a row releases heats the gas, except a photon's, which leaves, and that of G2, PEN_KR2 and SUPER, which goes to the electrons. KrF(B) and KrF(C) lie 80 cm^{-1} apart and are drawn as a pair of bars that every B and C row meets. The rows G6, VIB1, VIB2, VIB3 and VIB4 move nothing between levels and are not drawn.

LIGHT system map, basic network: energy flow through the Kr/F_2 kinetics

Reference point: 1 atm of Kr with 0.5 % F_2 at 300 K, P = 300 kW cm^{-3} for 1 μs so that E_{dep} = 0.30 J cm^{-3}, n_e = 10^{15} cm^{-3} held, run to 2 μs. Band width is proportional to the energy flow. Numbers are percent of E_{dep}, rounded to 0.1.



The notes' outputs as flows, which overlap: to Kr* 101.9 %, to Kr+ 27.1 %, to KrF(B) and KrF(C) 28.0 %, heat 42.9 %, elastic losses 8.9 %, emission 10.1 %. Of the KrF formed, 1.05×10^{17} cm^{-3}, 18.7 % radiates at 248 nm and 3.2 % in the C-A band, 50.5 % forms the trimer, and electrons, F_2 and Kr quench 19.0 %, 5.4 % and 3.2 %. B-C mixing carries 11.5 % of E_{dep} from B to C and 14.4 % back. At the end of the run T_g = 604 K and p = 2.03 atm, and F_2 fell below 10 % of the fill at 715 ns.

Generated from LIGHT v0 (0.1.0), basic network, by `docs/make_system_map_basic.py`. No number on this page is typed by hand. The rates and their sources are in `accuracy_overview.pdf`, Appendix A.

7 Where LIGHT is not yet reliable

- The trimer. KrF + 2Kr $\rightarrow$ Kr_2F^* takes 50.5 % of the KrF at the reference point. Its $(T_g/300)^{-3.32}$ law is Shui's calculation for 300 to 600 K, measured only between 315 and 420 K. At the notes' reference point the gas ends at 604 K.
- The drive. The notes' power drive feeds the electron gas directly and borrows the tail table made at e-beam states. LIGHT has no e-beam route of its own, and the Tahoma scores ran under a prototype beam drive that is not part of LIGHT's model.
- Fluorine. At the reference point F_2 falls below 10 % of the fill at 715 ns, inside the pulse, and LIGHT has no F + F recombination.
- Ions. With n_e held fixed and the ions group off, nothing recombines, so the ion store is bookkeeping.
- The C-A band centre near 275 nm is read from a figure to $\pm$10 nm.
- Hakuta's C-A decays and Burnham and Searles' τ_B remain unexplained conflicts.

A Reaction rates

Rates are the values at 300 K. The column T gives the law LIGHT applies to them. Electron-impact rows use $\langle \sigma v \rangle$ over the EEDF, and Section A.3 lists their Maxwellian values. Each row's full citation is in `v0/network.py`. The row ids are the ones the code and the system map use.

A.1 The basic network, 24 rows

row	reaction	rate	T	source
G1	$e^- + Kr \rightarrow e^- + Kr^*$	$\langle \sigma v \rangle$, Table A.3	T_e	Biagi v7.1, the whole 5s manifold at 9.915 eV
C1a	$e^- + Kr \rightarrow e^- + Kr^*$	$\langle \sigma v \rangle$, Table A.3	T_e	Biagi v7.1 5p group at 11.30 eV, prompt cascade to 5s
C1b	$e^- + Kr \rightarrow e^- + Kr^*$	$\langle \sigma v \rangle$, Table A.3	T_e	Biagi v7.1 d+p group at 11.998 eV, as C1a
C1c	$e^- + Kr \rightarrow e^- + Kr^*$	$\langle \sigma v \rangle$, Table A.3	T_e	Biagi v7.1 higher group at 12.75 eV, as C1a
C2	$e^- + Kr \rightarrow 2e^- + Kr^+$	$\langle \sigma v \rangle$, Table A.3	T_e	Biagi v7.1 ionization at 13.996 eV
G6	e^- + Kr, elastic	energy loss $(2m/M)\langle \sigma_m v \varepsilon \rangle$	T_e, T_g	Biagi v7.1 momentum transfer
G2	$2Kr^* \rightarrow Kr^+ + Kr + e^-$	5.0×10^{-10} cm^{3}s^{-1}		Kannari 1985 Table I, Penning block
G3	$Kr^* + F_2 \rightarrow KrF(B) + F$	4.73×10^{-10} cm^{3}s^{-1}		rate from Velazco, Kolts and Setser 1976 and Ueda 1984, KrF^* yield 0.90 from Velazco, Kolts and Setser 1976, B share from Yu and Setser 1990
G3C	$Kr^* + F_2 \rightarrow KrF(C) + F$	2.9×10^{-10} cm^{3}s^{-1}		as G3, with Yu and Setser's k_B/k_C of 1.6
G3X	$Kr^* + F_2 \rightarrow Kr + 2F$	8.47×10^{-11} cm^{3}s^{-1}		as G3, the KrF^*-free rest, assigned to Kr + 2F
G4	$KrF(B) \rightarrow Kr + F + h\nu$	1.43×10^{8} s^{-1}		Kannari 1985 Table I. The notes write about 4 ns
G4C	$KrF(C) \rightarrow Kr + F + h\nu$	2.0×10^{7} s^{-1}		ML joint fit. Hakuta 1990 measured 60 $\pm$ 10 ns
BC	$KrF(B) + Kr \rightarrow KrF(C) + Kr$	1.05×10^{-11} cm^{3}s^{-1}		ML joint fit. Hakuta 1990's 1.2×10^{-10} is held out
CB	$KrF(C) + Kr \rightarrow KrF(B) + Kr$	9.55×10^{-12} cm^{3}s^{-1}	$1/K_{eq}(T_g)$	detailed balance of BC, with the joint fit's K_{eq}
G5	$KrF(B) + 2Kr \rightarrow Kr_2F^* + Kr$	1.9×10^{-30} cm^{6}s^{-1}	$(T_g/300)^{-3.32}$	ML joint fit. T_g law from Shui's calculation

row	reaction	rate	T	source
G5C	KrF(C) + 2Kr → Kr_2F^* + Kr	3.79×10^{-31} cm^{6}s^{-1}	$(T_g/300)^{-3.32}$	ML joint fit
QE	KrF(B) + e^- → Kr + F + e^-	6.6×10^{-8} cm^{3}s^{-1}		Trainor and Jacob's measured product, re-read with this network's lifetime
QEC	KrF(C) + e^- → Kr + F + e^-	6.6×10^{-8} cm^{3}s^{-1}		as QE
QF2	KrF(B) + F_2 → Kr + 3F	4.75×10^{-10} cm^{3}s^{-1}		ML joint fit. Xu 1993 printed 3.0 ± 0.3×10^{-10} with fast mixing
QF2C	KrF(C) + F_2 → Kr + 3F	3.04×10^{-10} cm^{3}s^{-1}		ML joint fit
QKR	KrF(B) + Kr → 2Kr + F	1.0×10^{-14} cm^{3}s^{-1}		ML joint fit, at its lower bound
QKRC	KrF(C) + Kr → 2Kr + F	8.37×10^{-13} cm^{3}s^{-1}		ML joint fit
C3	Kr_2F^* → 2Kr + F + $h\nu$	4.5×10^6 s^{-1}		Xu, Gadomski and Setser 1993, 220 ± 20 ns
Q2F	Kr_2F^* + F_2 → 2Kr + F + F_2	1.2×10^{-10} cm^{3}s^{-1}		Xu 1993 Table I, 1.2 ± 0.2×10^{-10}

A.2 Optional groups on the basic network, 19 rows

The groups kr2, ions, electron, f2e are off at the reference point and switch on with `--add`. The first four Kr_2^* rows carry the basic network's measured values.

row	reaction	rate	T	source
AK8_Kr	Kr^* + 2Kr → Kr_2^* + Kr	1.7×10^{-32} cm^{6}s^{-1}	$(T_g/300)^{-0.64}$	Ueda 1984 Table V, 1.7 ± 0.2×10^{-32}. T_g law from Wieme and Lenaerts 1981
KR2_RAD	Kr_2^* → 2Kr + $h\nu$	3.7×10^6 s^{-1}		Morikawa 1989, 270 ± 5 ns, and five other measurements
NK7	Kr_2^* + F_2 → KrF(B) + Kr + F	1.05×10^{-10} cm^{3}s^{-1}		half of Quigley and Hughes 1978, 2.1 ± 0.2×10^{-10}. The even split is Kannari's
K2F_BR	Kr_2^* + F_2 → Kr_2F^* + F	1.05×10^{-10} cm^{3}s^{-1}		the other half of Quigley and Hughes 1978
K2F_F	Kr_2^* + F → KrF(B) + Kr	3.0×10^{-10} cm^{3}s^{-1}		Kannari 1985, an estimate. No measurement exists
PEN_KR2	2Kr_2^* → Kr_2^+ + 2Kr + e^-	5.0×10^{-10} cm^{3}s^{-1}		Kannari 1985 Table I
K15	Kr^+ + 2Kr → Kr_2^+ + Kr	2.5×10^{-31} cm^{6}s^{-1}	$(T_g/300)^{-0.75}$	Kannari 1985, Boichenko 2000
R11p	e^- + Kr_2^+ → Kr^* + Kr	$2.24\times10^{-7}\ T_e^{-1/2}$ cm^{3}s^{-1}	T_e	Kannari 1985 Table I, on a $T_e^{-1/2}$ template
KSTEP	e^- + Kr^* → Kr^+ + 2e^-	$1.56\times10^{-7}\ T_e^{0.71}\times e^{-4.09/T_e}$ cm^{3}s^{-1}	T_e	Chen R9 template, an estimate
SUPER	e^- + Kr^* → e^- + Kr	from G1 by detailed balance	T_e	detailed balance of G1, statistical weight ratio 1/12
ATT	e^- + F_2 → F^- + F	$\langle\sigma v\rangle$, Table A.3	T_e	SIGLO F_2 set
DIS_a3	e^- + F_2 → e^- + 2F	$\langle\sigma v\rangle$, Table A.3	T_e	SIGLO $a\,^3\Pi_u$, repulsive, so excitation dissociates
DIS_A1	e^- + F_2 → e^- + 2F	$\langle\sigma v\rangle$, Table A.3	T_e	SIGLO $A\,^1\Pi_u$, as DIS_a3
VIB1	e^- + F_2 → e^- + F_2	$\langle\sigma v\rangle$, Table A.3	T_e	SIGLO, v = 1
VIB2	e^- + F_2 → e^- + F_2	$\langle\sigma v\rangle$, Table A.3	T_e	SIGLO, v = 2
VIB3	e^- + F_2 → e^- + F_2	$\langle\sigma v\rangle$, Table A.3	T_e	SIGLO, v = 3
VIB4	e^- + F_2 → e^- + F_2	$\langle\sigma v\rangle$, Table A.3	T_e	SIGLO, v = 4
K26	Kr^+ + F^- → KrF(B)	2.95×10^{-6} cm^{3}s^{-1} at 1 atm, 300 K, unscreened	N, screening	Kannari 1985, Flannery turnover with Debye screening. No measurement with F^- exists
K108	Kr_2^+ + F^- → KrF(B) + Kr	2.95×10^{-6} cm^{3}s^{-1} at 1 atm, 300 K, unscreened	N, screening	as K26

A.3 Electron-impact rate coefficients

Maxwellian $\langle \sigma v \rangle$ in $\mathrm{cm^3 s^{-1}}$ at the T_e of each column. The last column is the factor the two-temperature tail applies at the reference point's mid-pulse state, T_e = 2.00 eV with the tail at 0.30 T_e. The rows below the rule give the energy an attached electron takes from the electron gas.

row	process	T_e (eV)					tail
		0.5	1	1.5	2	3	factor
G1	Kr, 5s manifold, 9.915 eV	1.24×10^{-17}	3.44×10^{-13}	1.11×10^{-11}	6.41×10^{-11}	3.78×10^{-10}	0.10
C1a	Kr, 5p group, 11.3 eV	5.1×10^{-19}	5.93×10^{-14}	2.97×10^{-12}	2.11×10^{-11}	1.48×10^{-10}	0.021
C1b	Kr, d+p group, 11.998 eV	2.5×10^{-19}	5.84×10^{-14}	3.77×10^{-12}	3.08×10^{-11}	2.53×10^{-10}	8.7×10^{-3}
C1c	Kr, higher group, 12.75 eV	9.15×10^{-21}	7.73×10^{-15}	8.8×10^{-13}	1.02×10^{-11}	1.29×10^{-10}	1.42×10^{-3}
C2	Kr, ionization, 13.996 eV	8.06×10^{-21}	1.57×10^{-14}	2.28×10^{-12}	2.94×10^{-11}	4.14×10^{-10}	5.84×10^{-4}
ATT	F_2, attachment	6.58×10^{-9}	3.51×10^{-9}	2.25×10^{-9}	1.61×10^{-9}	9.75×10^{-10}	1
DIS_a3	F_2, $a\,^3\Pi_u$, 3.16 eV	3.01×10^{-12}	1.13×10^{-10}	3.95×10^{-10}	7.37×10^{-10}	1.35×10^{-9}	1
DIS_A1	F_2, $A\,^1\Pi_u$, 4.34 eV	2.06×10^{-13}	1.93×10^{-11}	8.88×10^{-11}	1.9×10^{-10}	4.02×10^{-10}	1
VIB1	F_2, v = 1, 0.111 eV	1.77×10^{-9}	1.22×10^{-9}	8.72×10^{-10}	6.61×10^{-10}	4.27×10^{-10}	1
VIB2	F_2, v = 2, 0.219 eV	2.01×10^{-9}	1.49×10^{-9}	1.09×10^{-9}	8.37×10^{-10}	5.47×10^{-10}	1
VIB3	F_2, v = 3, 0.324 eV	1.21×10^{-9}	9.75×10^{-10}	7.35×10^{-10}	5.71×10^{-10}	3.78×10^{-10}	1
VIB4	F_2, v = 4, 0.421 eV	3.83×10^{-10}	3.27×10^{-10}	2.51×10^{-10}	1.96×10^{-10}	1.31×10^{-10}	1
ATT	energy an attached e^- takes, eV	0.37	0.46	0.53	0.59	0.72	
	$\frac{3}{2}T_e$, for comparison	0.75	1.50	2.25	3.00	4.50	

A.4 Where v0, the notes' list as written, differs

The v0 network keeps the notes' six reactions with REBUILD's values, and its optional groups, quench, kr2, ions, electron, bc, f2e, are REBUILD's. These rows differ from the basic network or exist only in v0. The rows ending in _C are the bc group's copies for KrF(C).

row	reaction	rate	T	source
G3	$Kr^* + F_2 \rightarrow KrF(B) + F$	8.1×10^{-10} $\mathrm{cm^3 s^{-1}}$		Kannari 1985 Table I, all to B. The report it credits gives 7.2×10^{-10}
G5	$KrF(B) + 2Kr \rightarrow Kr_2F^* + Kr$	6.0×10^{-31} $\mathrm{cm^6 s^{-1}}$	$(T_g/300)^{-3.32}$	Xu, Gadomski and Setser 1993 Table II, $6.0 \pm 2.0\times10^{-31}$
G5_C	$KrF(C) + 2Kr \rightarrow Kr_2F^* + Kr$	6.0×10^{-31} $\mathrm{cm^6 s^{-1}}$	$(T_g/300)^{-3.32}$	as G5, applied to KrF(C)
AK8_Kr	$Kr^* + 2Kr \rightarrow Kr_2^* + Kr$	2.7×10^{-32} $\mathrm{cm^6 s^{-1}}$		REBUILD's estimate, Kannari's Ar-partner rate given to Kr
KR2_RAD	$Kr_2^* \rightarrow 2Kr + h\nu$	8.0×10^{7} $\mathrm{s^{-1}}$		Kannari 1985 Table I, 8.0×10^7 per second
NK7	$Kr_2^* + F_2 \rightarrow KrF(B) + Kr + F$	3.0×10^{-10} $\mathrm{cm^3 s^{-1}}$		Kannari 1985 Table I
K2F_BR	$Kr_2^* + F_2 \rightarrow Kr_2F^* + F$	3.0×10^{-10} $\mathrm{cm^3 s^{-1}}$		Kannari 1985 Table I
K33	$KrF(B) + Kr \rightarrow 2Kr + F$	4.0×10^{-12} $\mathrm{cm^3 s^{-1}}$		Xu, Gadomski and Setser 1993, $4.0 \pm 0.5\times10^{-12}$
K33_C	$KrF(C) + Kr \rightarrow 2Kr + F$	4.0×10^{-12} $\mathrm{cm^3 s^{-1}}$		as K33, applied to KrF(C)
K32	$KrF(B) + F_2 \rightarrow Kr + 3F$	3.0×10^{-10} $\mathrm{cm^3 s^{-1}}$		Xu, Gadomski and Setser 1993, $3.0 \pm 0.3\times10^{-10}$
K32_C	$KrF(C) + F_2 \rightarrow Kr + 3F$	3.0×10^{-10} $\mathrm{cm^3 s^{-1}}$		as K32, applied to KrF(C)
K67	$KrF(B) + e^- \rightarrow Kr + F + e^-$	4.5×10^{-8} $\mathrm{cm^3 s^{-1}}$		Kushner, Hanson and Schneider 1989
K67_C	$KrF(C) + e^- \rightarrow Kr + F + e^-$	4.5×10^{-8} $\mathrm{cm^3 s^{-1}}$		as K67, applied to KrF(C)
K125	$Kr_2F^* + F_2 \rightarrow 2Kr + F + F_2$	1.5×10^{-10} $\mathrm{cm^3 s^{-1}}$		Kannari 1985 Table I
K_2Fe	$Kr_2F^* + e^- \rightarrow 2Kr + F + e^-$	1.0×10^{-7} $\mathrm{cm^3 s^{-1}}$		Kannari 1985, estimated there
BC_Kr	$KrF(B) + Kr \rightarrow KrF(C) + Kr$	5.0×10^{-10} $\mathrm{cm^3 s^{-1}}$		Kannari's B-C mixing with a Kr partner, an estimate
CB_Kr	$KrF(C) + Kr \rightarrow KrF(B) + Kr$	3.41×10^{-10} $\mathrm{cm^3 s^{-1}}$		detailed balance of BC_Kr at 300 K
URAD_C	$KrF(C) \rightarrow Kr + F + h\nu$	1.33×10^{7} $\mathrm{s^{-1}}$		Kannari 1985 Table I, 1.33×10^7 per second

A.5 Formation energies

Energies above ground-state Kr and F_2 at rest. Every reaction's energy release is the difference of these numbers, so no row can make or lose energy. c_v is the heat capacity per particle in units of k_B.

species	E_{form} (eV)	c_v	what sets it
Kr	0	1.5	ground state
F_2	0	2.5	ground state
F	0.8	1.5	half the F_2 bond, D_0 = 1.60 eV
Kr^*	9.915	1.5	Biagi's 5s threshold, the 5s[3/2]2 level
Kr^+	13.996	1.5	Biagi's ionization threshold
KrF(B)	6.15	2.5	F + the 248-nm quantum + 0.35 eV of X-state repulsion
KrF(C)	6.1401	2.5	80 cm^{-1} below B
Kr_2F^*	4.1	3	F + about 3.0 eV of 415-nm band + about 0.3 eV
Kr_2^*	9.2	2.5	the 147-nm excimer, an estimated well
Kr_2^+	12.846	2.5	IP(Kr) less $D_0(Kr_2^+)$ = 1.15 eV
F^-	−2.6	1.5	E_{form}(F) less the electron affinity, 3.40 eV

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